

Ticket 09 Study Report: Continuation Premium and Boundary Extraction

SURF2026 American Option Risk Surfaces

Codex-assisted implementation report

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Abstract

Ticket 09 is the first project ticket where free-boundary extraction is intentionally in scope. It uses existing American CN/PSOR outputs and computes the continuation premium $U - \Phi$, where U is the American option value and Φ is the payoff. The extraction layer then looks for option-type-aware transitions between exercise-like and continuation-like regions. This report explains the method for beginner researchers who know Python and data work but are still learning American option free-boundary diagnostics.

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1 Role of Ticket 09 in the Whole Project

Tickets 01 through 04 built the pricing machinery: payoff functions, Black-Scholes utilities, finite-difference grids, European Crank–Nicolson validation, and the American CN/PSOR core. Tickets 05 through 07 validated American put, no-dividend American call, and dividend-paying American call behavior. Ticket 08 then checked obstacle and complementarity diagnostics without extracting a boundary.

Ticket 09 is the natural next layer. It takes the stored American value grid and asks:

- Where does the value equal payoff, up to a stated threshold?
- Where does the value become strictly greater than payoff?
- Can we record the transition as an approximate boundary without pretending it is exact?

The output is a boundary-extraction diagnostic, not a new pricing algorithm. The solver core is unchanged.

2 Theory and Methodology Connection

The documented source basis for Ticket 09 is internal:

- the planning report, which places boundary extraction after solver validation and complementarity diagnostics;
- the literature map, which emphasizes that boundary extraction should use continuation premium rather than visual guessing;
- the formulation note, especially the sections on continuation premium and free-boundary extraction;
- the solver validation plan, especially its boundary extraction validation guidance;
- the Ticket 04 through Ticket 08 implementation reports and tests.

This reference integration pass adds verified citations for the external methodology behind the project documents. The approximate free-boundary and finite-difference context is connected to Brennan-Schwartz and computational-finance PDE background (Brennan and Schwartz, 1977; Wilmott et al., 1995), while complementarity terminology is connected to standard LCP methodology (Cottle et al., 2009).

3 Continuation Premium $U - \Phi$

For an American option, the value $U(S, \tau)$ should not fall below the immediate exercise payoff $\Phi(S)$. Define the continuation premium:

$$P_{\text{cont}}(S, \tau) = U(S, \tau) - \Phi(S).$$

In plain language, this is the extra value from holding the option instead of exercising immediately. The interpretation is:

- $P_{\text{cont}} \approx 0$: the value is close to payoff, so the node is exercise-like;
- $P_{\text{cont}} > 0$: waiting has extra value, so the node is continuation-like;
- the free boundary is approximated where the premium crosses from near zero to positive.

The word “approximately” matters. A finite-difference grid has spacing error, PSOR tolerance, and a discrete threshold. Ticket 09 therefore reports the threshold and no-boundary reasons.

4 Why Boundary Extraction Comes After Diagnostics

Boundary extraction is more fragile than a price comparison. A price can look reasonable while the obstacle or complementarity conditions fail. That is why Ticket 08 came first: it checked that the representative American put and dividend-paying American call results satisfy the LCP diagnostics at interior nodes.

Ticket 09 uses that groundwork. It does not re-solve the option and does not recompute LCP residuals. It reads the stored value grid, computes $U - \Phi$, and extracts approximate transition points. If future diagnostics fail, boundary estimates should be treated with suspicion.

5 Put Search Direction

For an American put, early exercise is expected at low spot prices. When the stock is very low, the put is deep in the money, and immediate exercise can be attractive. Therefore the search rule is:

1. move from low S toward high S ;
2. identify the low- S exercise-like run where $P_{\text{cont}} \leq \epsilon$;
3. identify the next continuation-like run where $P_{\text{cont}} > \epsilon$;
4. interpolate the threshold crossing between the last exercise-like node and first continuation-like node.

The far high- S tail of a put can also have very small option value and therefore very small premium. Ticket 09 does not treat that far tail as the exercise boundary. It uses the option-type search direction to find the economically relevant low-spot transition.

6 Dividend-Call Search Direction

For a dividend-paying American call, early exercise can become meaningful at high spot prices. Exercising early sacrifices optionality, but it may let the holder own the stock before dividends reduce its value. Therefore the search rule is:

1. move from high S toward lower S ;
2. identify the high- S exercise-like run where $P_{\text{cont}} \leq \epsilon$;
3. identify the adjacent continuation-like run below it;
4. interpolate the threshold crossing between the continuation-like and exercise-like nodes.

The low- S tail of an out-of-the-money call can also have tiny value and tiny premium. Ticket 09 ignores that low-spot tail for call boundary extraction, because the dividend-call exercise region of interest is the high-spot side.

7 Threshold and Interpolation Choices

Ticket 09 uses the default threshold

$$\epsilon = 10^{-6}.$$

The rule is:

$$\text{exercise-like if } P_{\text{cont}}(S_i, \tau) \leq \epsilon, \quad \text{continuation-like if } P_{\text{cont}}(S_i, \tau) > \epsilon.$$

If a clean transition exists between adjacent nodes S_i and S_{i+1} , Ticket 09 uses linear interpolation:

$$S_f = S_i + \frac{\epsilon - P_i}{P_{i+1} - P_i} (S_{i+1} - S_i),$$

where $P_i = P_{\text{cont}}(S_i, \tau)$. This is a practical grid diagnostic, not an exact analytical boundary formula.

Boundary extraction decisions use interior spot nodes to reduce imposed-boundary artifacts. Full-grid premium profiles are still stored and plotted for study purposes.

8 No-Boundary Metadata

Ticket 09 does not silently return a missing value. Each time row records a reason when no boundary is extracted:

Reason	Meaning
maturity_row_ambiguous	At $\tau = 0$, value equals payoff everywhere, so the whole row is exercise-like by construction.
all_continuation_like	No interior node has premium below the threshold.
all_exercise_like	Every interior node is threshold-exercise-like.
expected_exercise_side_absent	The option-type side where exercise is expected does not show threshold-exercise behavior.
no_clean_transition	The observed pattern is not a stable transition under the extraction rule.
insufficient_interior_nodes	There are not enough interior nodes to bracket a transition.

9 Continuation Premium Versus American-Minus-European Value

Continuation premium is not the same as American-minus-European value.

$$P_{\text{cont}} = U_{\text{American}} - \Phi.$$

American-minus-European value is:

$$U_{\text{American}} - U_{\text{European}}.$$

The first quantity compares the American value to immediate exercise. It is the right quantity for obstacle and boundary extraction. The second compares the American contract to a European contract. It is useful for validation, especially in Tickets 05 through 07, but it is not the boundary-extraction object.

10 Implementation Design Decisions

Decision	Reason
Create a diagnostics module	Boundary extraction is a diagnostic layer on solver outputs, not part of the solver core.
Use Ticket 04 result objects	The stored value grid, payoff, spot grid, and tau grid already contain the needed data.
Keep solver files unchanged	Ticket 09 should not introduce a new pricing algorithm or alter CN/PSOR logic.
Use full grids for profiles	Full profiles help students see the premium shape across the domain.
Use interior nodes for extraction	Interior extraction avoids treating imposed boundary values as solved boundary locations.
Record no-boundary reasons	Missing boundaries are informative, especially for the no-dividend call control.
Include no-dividend control	It checks that the extractor does not force a high-spot call boundary when $q = 0$.

11 Code-Level Function Explanations

Function or class	Role
BoundaryPoint	Stores one time-level boundary estimate or no-boundary reason.
BoundaryCurve	Stores all time-level boundary points plus value, payoff, and premium grids.
BoundaryExtractionSummary	Stores whole-curve counts, boundary ranges, and status.
continuation_premium	Computes $U - \Phi$ for one row or a full value grid.
linear_interpolate_threshold_crossing	Estimates the threshold-crossing spot between adjacent nodes.
extract_boundary_at_time	Applies the option-type search rule to one premium row.
extract_boundary_curve	Applies boundary extraction over all time-to-maturity rows.
summarize_boundary_curve	Converts all boundary points into summary counts and ranges.
selected_time_profile_rows	Exports full-grid premium rows at selected time fractions for study and plotting.
run_boundary_cases	Runs the three Ticket 09 experiment cases and prepares CSV/figure artifacts.

12 Extraction Cases and Parameters

Case	Option	K	T	r	q	σ	M	N
Put medium	put	1.0	1.0	0.05	0.02	0.20	120	120
Dividend call	call	1.0	1.0	0.05	0.08	0.20	120	120
No-dividend call	call	1.0	1.0	0.05	0.00	0.20	120	120

All cases use $S_{\max} = 4.0$ and threshold 10^{-6} . The no-dividend call control is not a repeat of Ticket 06 price validation. It is a boundary-extraction control.

13 Boundary Result Tables and Interpretation

13.1 Boundary Summary

Case	Direction	Found rows	No-boundary rows	Min boundary	Max boundary
Put medium	low-to-high	120	1	0.766754	0.934082
Dividend call	high-to-low	120	1	1.066165	1.266630
No-dividend call	high-to-low	0	121	n/a	n/a

The put and dividend-call cases find boundaries for every positive- τ row. The maturity row is marked ambiguous because $U = \Phi$ everywhere at $\tau = 0$. The no-dividend call control finds no boundary, which is the desired behavior for a control case.

13.2 No-Boundary Reason Counts

Case	Maturity	All cont.	Expected side absent	No clean transition
Put medium	1	0	0	0
Dividend call	1	0	0	0
No-dividend call	1	0	120	0

The control case reports `expected_exercise_side_absent` for positive time rows. In plain language, the high- S side of the no-dividend call does not show an exercise-like transition.

13.3 Selected By-Time Rows

Case	Step	τ	Found	Boundary spot	Reason
put	0	0.00000	false	n/a	maturity ambiguous
put	1	0.00833	true	0.934082	—
put	60	0.50000	true	0.800004	—
put	120	1.00000	true	0.766754	—

Case	Step	τ	Found	Boundary spot	Reason
dividend call	0	0.00000	false	n/a	maturity ambiguous
dividend call	1	0.00833	true	1.066165	—
dividend call	60	0.50000	true	1.200001	—
dividend call	120	1.00000	true	1.266630	—
no-dividend call	1	0.00833	false	n/a	expected side absent
no-dividend call	120	1.00000	false	n/a	expected side absent

These rows are selected for readability. The full by-time CSV contains 363 rows: three cases times 121 time rows.

14 Figure Interpretation

Figure 1 plots the American put boundary over time-to-maturity. The boundary is below the strike and moves gradually on the tested grid.

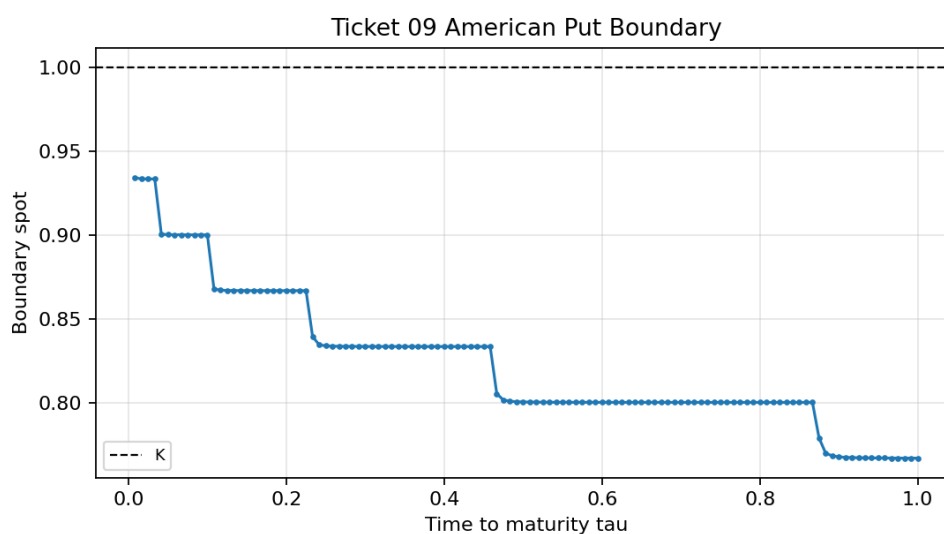


Figure 1: Ticket 09 American put boundary curve extracted from continuation premium.

Figure 2 plots the dividend-paying American call boundary. It appears above the strike in this representative $q = 0.08$ case.

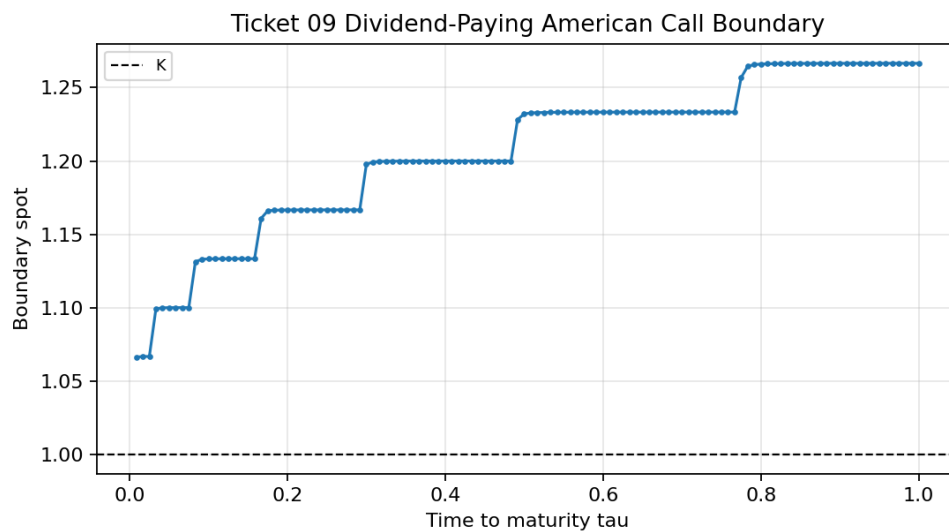


Figure 2: Ticket 09 dividend-paying American call boundary curve extracted from continuation premium.

Figure 3 shows selected premium profiles. These curves help students see why the boundary is a threshold-crossing diagnostic rather than a visual guess.

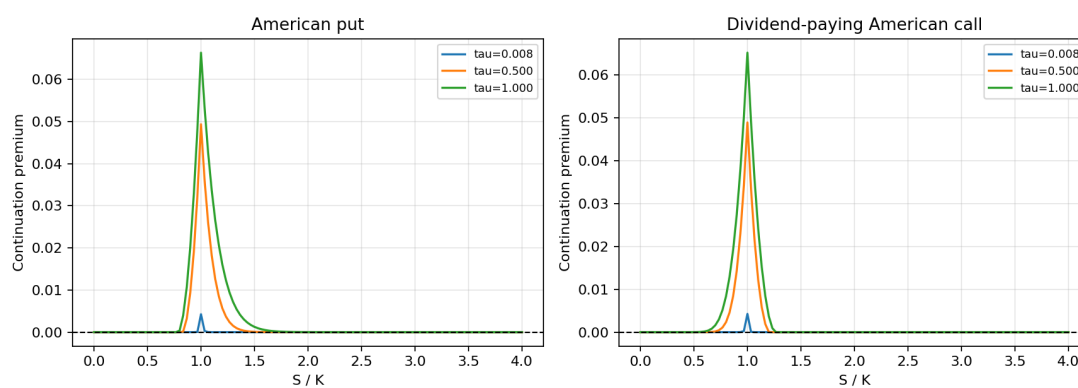


Figure 3: Ticket 09 continuation-premium profiles at selected time-to-maturity levels.

Figure 4 records whether a boundary was found at each time row. The no-dividend call control stays at “not found” for positive time rows.

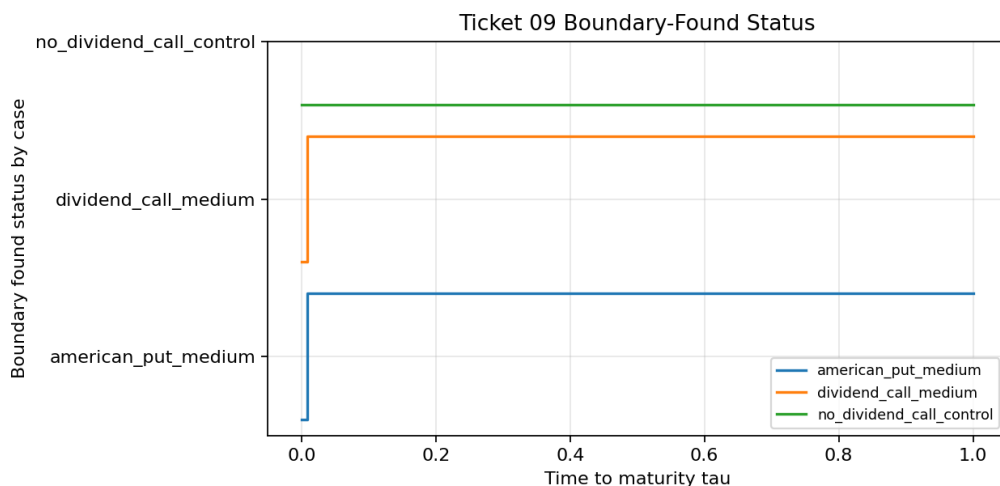


Figure 4: Ticket 09 boundary-found status by case and time-to-maturity row.

15 Testing Strategy

Test category	What it proves	What it does not prove
Premium calculation	$U - \Phi$ is computed with correct broadcasting.	Boundary accuracy.
Synthetic put boundary	Low-spot put search direction works on a controlled transition.	Real put convergence.
Synthetic call boundary	High-spot dividend-call search direction works on a controlled transition.	Real call convergence.
No-boundary metadata	Missing boundaries are explained rather than hidden.	That every future missing boundary is harmless.
Interpolation	The threshold crossing formula is predictable.	Higher-order boundary accuracy.
Real put and call smoke cases	Existing solver outputs produce boundary rows and nonnegative premiums.	Full grid/domain sensitivity.
No-dividend control	The extractor does not force a high-spot call boundary for $q = 0$.	Ticket 06 theorem validation.
CSV and figure artifacts	Reproducible tables and useful figures are produced.	Scientific proof by plotting.
Scope and docstrings	Ticket 09 files identify the ticket and avoid Greeks, stress maps, datasets, and neural work.	Later-ticket functionality.

16 Test Results and Interpretation

The Ticket 09 implementation followed a test-first workflow. The first red run failed because `src/american_risk_surfaces/diagnostics/boundary.py` and `experiments/06_boundary_extraction.py` did not exist yet. After implementation, the targeted Ticket 09 test command reported:

```
Ran 17 tests in 0.611s
OK
```

The full test command reported:

```
Ran 104 tests in 1.824s
OK
```

This means the Ticket 09 tests passed together with all earlier Ticket 01–08 tests. It does not mean the extracted boundaries are exact analytical free boundaries.

17 Implementation Pitfalls and Debugging Notes

Recorded item or pitfall	Symptom	Fix or guard	Beginner lesson
Expected TDD red run	New tests failed because the Ticket 09 module and experiment did not exist.	Add the planned module and experiment.	Red tests confirm new behavior is actually being tested.
Malformed targeted unittest command	The dotted test-module form could not import because <code>tests/</code> is not a package.	Use unittest discovery with the tests directory and filename pattern.	Match test commands to repo layout.
Overly strict transition rule	Real rows looked like exercise-like, continuation-like, exercise-like because far tails can have tiny premium.	Use option-type search direction and require a minimum continuation block near the expected exercise side.	Boundary extraction needs financial direction, not only boolean pattern matching.
Confusing premium types	American-minus-European value is tempting to plot.	Use $U - \Phi$ for boundary extraction.	Boundary extraction uses payoff comparison, not European comparison.
Forcing a control boundary	A no-dividend call may have tiny low-spot premium.	Search high-to-low for calls and report expected-side absence for $q = 0$.	Negative controls are valuable.

Recorded item or pitfall	Symptom	Fix or guard	Beginner lesson
Overclaiming exactness	A smooth-looking curve can feel authoritative.	Report threshold, interpolation, and limitations.	Numerical boundaries are estimates with metadata.

Future coding/test reports should continue to include implementation pitfalls when failed tests, debugging discoveries, or workflow corrections occur.

18 Limitations

Not included in Ticket 09	Reason
New pricing algorithm	Ticket 09 reads existing American CN/PSOR outputs only.
Greeks	Delta and Gamma belong to Ticket 10.
Greek masks near boundary	Masking depends on later Greek diagnostics.
Stress maps	Regime sweeps are later validation work.
Datasets and neural networks	Labels and models require a more mature validated solver pipeline.
Exact free-boundary accuracy claim	The current output is threshold-based and grid-dependent.
Full grid/domain sensitivity	This ticket uses representative $M = N = 120$ cases only.

19 Lessons Learned / Study Notes

The first lesson is that a free boundary is not drawn by eye. It should come from a defined quantity, here the continuation premium $U - \Phi$.

The second lesson is that search direction matters. Put boundaries and dividend-call boundaries live on different sides of the spot domain.

The third lesson is that missing boundaries are data, not failure by default. The no-dividend American call control should not have a forced high-spot exercise boundary.

The fourth lesson is that thresholds must be reported. A boundary extracted with $\epsilon = 10^{-6}$ is not the same claim as a boundary extracted with a much larger threshold.

The fifth lesson is to keep later tickets separate. Boundary extraction is now allowed, but Greeks, stress maps, datasets, and neural-network labels are still out of scope.

20 Link to Ticket 10

Ticket 10 should compute Delta and Gamma diagnostics. It can use the Ticket 09 boundary curves to define careful reporting bands near the free boundary, but it should not treat the current boundary estimates as exact analytical truth. Greek diagnostics should remain honest about payoff kinks, threshold effects, and boundary-near numerical instability.

21 Reproducibility Record

Item	Record
Boundary module	src/american_risk_surfaces/diagnostics/boundary.py.
Experiment	experiments/06_boundary_extraction.py.
Tests	tests/test_boundary_extraction.py.
Summary CSV	results/01_solver_validation/tables/ticket_09_boundary_extraction_summary.csv.
By-time CSV	results/01_solver_validation/tables/ticket_09_boundary_by_time.csv.
Selected-times CSV	results/01_solver_validation/tables/ticket_09_boundary_selected_times.csv.
Figures	results/01_solver_validation/figures/ticket_09_american_put_boundary_curve.png; results/01_solver_validation/figures/ticket_09_dividend_call_boundary_curve.png; results/01_solver_validation/figures/ticket_09_continuation_premium_profiles.png; results/01_solver_validation/figures/ticket_09_boundary_found_status.png.
Report source	reports/03_solver/tickets/ticket_09_continuation_premium_boundary_extraction.tex.
Report PDF	reports/03_solver/tickets/ticket_09_continuation_premium_boundary_extraction.pdf.

Commands used:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
tmp/matplotlib-cache FC_CACHEDIR=/private/tmp/fontconfig-cache python3
experiments/06_boundary_extraction.py
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
tmp/matplotlib-cache FC_CACHEDIR=/private/tmp/fontconfig-cache python3 -m
unittest discover -s tests
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src python3 -m compileall
-q src tests experiments
```

References

- Michael J. Brennan and Eduardo S. Schwartz. The valuation of american put options. *The Journal of Finance*, 32(2):449, 1977. doi: 10.2307/2326779.
- Richard W. Cottle, Jong-Shi Pang, and Richard E. Stone. *The Linear Complementarity Problem*. Society for Industrial and Applied Mathematics, 2009. ISBN 9780898716863. doi: 10.1137/1.9780898719000.
- Paul Wilmott, Sam Howison, and Jeff Dewynne. *The Mathematics of Financial Derivatives*. Cambridge University Press, 1995. doi: 10.1017/CBO9780511812545.